JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Quadratic Equation and Inequalities

JEE Main Mathematics · Previous Year Questions with Answer Key

198	157	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	160 questions
Numerical	38 questions
Answer key	at the end

1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

MCQ (Single Correct Answer) 160 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Let $\lim\limits_{x\to 2}\frac{(\tan(x-2))\left(rx^2+(p-2)x-2p\right)}{(x-2)^2}=5$ for some $r, p \in \boldsymbol{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2-px+q=0$ lie in $(0,2)$, be the interval $(\alpha,\beta]$, then $4(\alpha+\beta)$ equals :

(A) 11 (B) 13 (C) 17 (D) 21

Q2. JEE Main 2026 (Online) 4th April Evening Shift

If the quadratic equation $(\lambda+2)x^2-3\lambda x+4\lambda=0, \lambda\neq -2$, has two positive roots, then the number of possible integral values of λ is :

(A) 1 (B) 2 (C) 3 (D) 4

Q3. JEE Main 2026 (Online) 4th April Morning Shift

If the set of all solutions of $\left|x^2+x-9\right|=|x|+\left|x^2-9\right|$ is $[\alpha,\beta]\cup[\gamma,\infty)$, then ($\alpha^2+\beta^2+\gamma^2$) is equal to:

(A) 9 (B) 18 (C) 36 (D) 72

Q4. JEE Main 2026 (Online) 2nd April Evening Shift

Let α, β be the roots of the equation $x^2-3x+r=0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2+3x+r=0$.

If the roots of the equation $x^2+6x=m$ are $2\alpha+\beta+2r$ and $\alpha-2\beta-\frac{r}{2}$, then m is equal to :

(A) -135 (B) -567 (C) 135 (D) 567

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

Let $\alpha, \alpha+2, \alpha\in\mathbf{Z}$, be the roots of the quadratic equation $x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\ldots.+(x+n-1)(x+n+1)=4n$ for some $n\in\mathbf{N}$. Then $n+\alpha$ is equal to :

(A) 0 (B) 1 (C) 2 (D) 3

Q6. JEE Main 2026 (Online) 28th January Morning Shift

If α, β, where $\alpha<\beta$, are the roots of the equation $\lambda x^2-(\lambda+3)x+3=0$ such that $\frac{1}{\alpha}-\frac{1}{\beta}=\frac{1}{3}$, then the sum of all possible values of λ is

(A) 2 (B) 6 (C) 8 (D) 4

Q7. JEE Main 2026 (Online) 28th January Morning Shift

Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S , which are divisible by $x^2 + 2$, is

(A) 6 (B) 120 (C) 20 (D) 10

Q8. JEE Main 2026 (Online) 24th January Evening Shift

The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

(A) 7 (B) 3 (C) 4 (D) 9

Q9. JEE Main 2026 (Online) 24th January Morning Shift

The number of the real solutions of the equation: $x|x + 3| + |x - 1| - 2 = 0$ is

(A) 3 (B) 2 (C) 5 (D) 4

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

If α and $\beta(\alpha < \beta)$ are the roots of the equation
$(-2 + \sqrt{3})(|\sqrt{x} - 3|) + (x - 6\sqrt{x}) + (9 - 2\sqrt{3}) = 0, x \geqslant 0$, then $\sqrt{\frac{\beta}{\alpha}} + \sqrt{\alpha\beta}$ is equal to :

(A) 8 (B) 10 (C) 9 (D) 11

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

(A) 36 days (B) 24 days (C) 30 days (D) 42 days

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leqslant |\beta - \alpha| \leqslant \frac{3}{2}$, then the sum of all possible values of λ is :

(A) 3 (B) 6 (C) 1 (D) 4

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

The number of distinct real solutions of the equation $x|x + 4| + 3|x + 2| + 10 = 0$ is

(A) 2 (B) 3 (C) 0 (D) 1

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Let α and β be the roots of the equation $x^2 + 2ax + (3a + 10) = 0$ such that $\alpha < 1 < \beta$. Then the set of all possible values of a is :

(A) $\left(-\infty, \frac{-11}{5}\right) \cup (5, \infty)$

(B) $\left(-\infty, \frac{-11}{5}\right)$

(C) $(-\infty, -3)$

(D) $(-\infty, -2) \cup (5, \infty)$

Q15. JEE Main 2026 (Online) 21st January Morning Shift

The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is :

(A) 3

(B) 1

(C) 4

(D) 5

Q16. JEE Main 2025 (Online) 8th April Evening Shift

The sum of the squares of the roots of $|x-2|^2 + |x-2| - 2 = 0$ and the squares of the roots of $x^2 - 2|x-3| - 5 = 0$, is

(A) 24

(B) 26

(C) 36

(D) 30

Q17. JEE Main 2025 (Online) 7th April Evening Shift

The number of real roots of the equation $x|x-2| + 3|x-3| + 1 = 0$ is :

(A) 4

(B) 3

(C) 2

(D) 1

Q18. JEE Main 2025 (Online) 7th April Morning Shift

Let the set of all values of $p \in \mathbb{R}$, for which both the roots of the equation $x^2 - (p+2)x + (2p+9) = 0$ are negative real numbers, be the interval $(\alpha, \beta]$. Then $\beta - 2\alpha$ is equal to

(A) 5

(B) 0

(C) 20

(D) 9

Q19. JEE Main 2025 (Online) 4th April Morning Shift

Consider the equation $x^2 + 4x - n = 0$, where $n \in [20, 100]$ is a natural number. Then the number of all distinct values of n, for which the given equation has integral roots, is equal to

(A) 6

(B) 5

(C) 8

(D) 7

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

Let the equation $x(x+2)(12-k) = 2$ have equal roots. Then the distance of the point $\left(k, \frac{k}{2}\right)$ from the line $3x + 4y + 5 = 0$ is

(A) 15

(B) 12

(C) $5\sqrt{3}$

(D) $15\sqrt{5}$

Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Let α and β be the roots of $x^2+\sqrt{3}x-16=0$, and γ and δ be the roots of $x^2+3x-1=0$. If $P_n=\alpha^n+\beta^n$ and $Q_n=\gamma^n+\delta^n$, then $\frac{P_{25}+\sqrt{3}P_{24}}{2P_{23}}+\frac{Q_{25}-Q_{23}}{Q_{24}}$ is equal to

(A) 4 (B) 3 (C) 5 (D) 7

Q22. JEE Main 2025 (Online) 2nd April Morning Shift

Let $\mathrm{P_n}=\alpha^{\mathrm{n}}+\beta^{\mathrm{n}}, \mathrm{n}\in \mathrm{N}$. If $\mathrm{P_{10}}=123, \mathrm{P_9}=76, \mathrm{P_8}=47$ and $\mathrm{P_1}=1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is :

(A) $x^2+x-1=0$ (B) $x^2-x+1=0$ (C) $x^2+x+1=0$ (D) $x^2-x-1=0$

Q23. JEE Main 2025 (Online) 29th January Evening Shift

If the set of all $a\in\mathbf{R}$, for which the equation $2x^2+(a-5)x+15=3a$ has no real root, is the interval (α,β), and $X=\{x\in Z;\alpha<x<\beta\}$, then $\sum\limits_{x\in X}x^2$ is equal to:

(A) 2139 (B) 2119 (C) 2109 (D) 2129

Q24. JEE Main 2025 (Online) 29th January Morning Shift

The number of solutions of the equation

$\left(\frac{9}{x}-\frac{9}{\sqrt{x}}+2\right)\left(\frac{2}{x}-\frac{7}{\sqrt{x}}+3\right)=0$ is :

(A) 3 (B) 2 (C) 1 (D) 4

Q25. JEE Main 2025 (Online) 28th January Evening Shift

Let $f:\mathbf{R}-\{0\}\to(-\infty,1)$ be a polynomial of degree 2 , satisfying $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If $f(\mathrm{K})=-2\,\mathrm{K}$, then the sum of squares of all possible values of K is :

(A) 9 (B) 1 (C) 6 (D) 7

Q26. JEE Main 2025 (Online) 28th January Morning Shift

The sum, of the squares of all the roots of the equation $x^2+|2x-3|-4=0$, is

(A) $6(2-\sqrt{2})$ (B) $3(3-\sqrt{2})$ (C) $3(2-\sqrt{2})$ (D) $6(3-\sqrt{2})$

Q27. JEE Main 2025 (Online) 24th January Evening Shift

The number of real solution(s) of the equation $x^2+3x+2=\min\{|x-3|,|x+2|\}$ is :

(A) 2 (B) 3 (C) 1 (D) 0

Q28. JEE Main 2025 (Online) 24th January Morning Shift

The product of all the rational roots of the equation $\left(x^2-9x+11\right)^2-(x-4)(x-5)=3$, is equal to

(A) 7 (B) 21 (C) 28 (D) 14

Q29. JEE Main 2025 (Online) 22nd January Evening Shift

Let α_θ and β_θ be the distinct roots of $2x^2+(\cos\theta)x-1=0, \theta\in(0,2\pi)$. If m and M are the minimum and the maximum values of $\alpha_\theta^4+\beta_\theta^4$, then $16(M+m)$ equals :

(A) 27 (B) 17 (C) 25 (D) 24

Q30. JEE Main 2024 (Online) 9th April Evening Shift

Let $\alpha,\beta;\alpha>\beta$, be the roots of the equation $x^2-\sqrt{2}x-\sqrt{3}=0$. Let $\mathrm{P}_n=\alpha^n-\beta^n, n\in\mathrm{N}$. Then $(11\sqrt{3}-10\sqrt{2})\mathrm{P}_{10}+(11\sqrt{2}+10)\mathrm{P}_{11}-11\mathrm{P}_{12}$ is equal to

(A) $10\sqrt{3}\mathrm{P}_9$ (B) $11\sqrt{3}\mathrm{P}_9$ (C) $11\sqrt{2}\mathrm{P}_9$ (D) $10\sqrt{2}\mathrm{P}_9$

Q31. JEE Main 2024 (Online) 9th April Morning Shift

Let α,β be the roots of the equation $x^2+2\sqrt{2}x-1=0$. The quadratic equation, whose roots are $\alpha^4+\beta^4$ and $\frac{1}{10}(\alpha^6+\beta^6)$, is:

(A) $x^2-180x+9506=0$ (B) $x^2-195x+9506=0$

(C) $x^2-190x+9466=0$ (D) $x^2-195x+9466=0$

Q32. JEE Main 2024 (Online) 8th April Morning Shift

The sum of all the solutions of the equation $(8)^{2x}-16\cdot(8)^x+48=0$ is :

(A) $1+\log_8(6)$ (B) $1+\log_6(8)$ (C) $\log_8(6)$ (D) $\log_8(4)$

Q33. JEE Main 2024 (Online) 6th April Morning Shift

Let α,β be the distinct roots of the equation $x^2-\left(t^2-5t+6\right)x+1=0, t\in\mathbb{R}$ and $a_n=\alpha^n+\beta^n$. Then the minimum value of $\frac{a_{2023}+a_{2025}}{a_{2024}}$ is

(A) $-1/2$ (B) $-1/4$ (C) $1/4$ (D) $1/2$

Q34. JEE Main 2024 (Online) 4th April Morning Shift

If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is :

(A) $x^2+8x+12=0$ (B) $2x^2+11x+12=0$

(C) $4x^2+14x+12=0$ (D) $x^2+10x+16=0$

Q35. JEE Main 2024 (Online) 1st February Evening Shift

Let α and β be the roots of the equation $px^2 + qx - r = 0$, where $p \neq 0$. If p, q and r be the consecutive terms of a non constant G.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$, then the value of $(\alpha - \beta)^2$ is :

(A) 8 (B) 9 (C) $\frac{20}{3}$ (D) $\frac{80}{9}$

Q36. JEE Main 2024 (Online) 1st February Morning Shift

Let $\mathbf{S} = \left\{x \in \mathbf{R} : (\sqrt{3}+\sqrt{2})^x + (\sqrt{3}-\sqrt{2})^x = 10\right\}$. Then the number of elements in S is :

(A) 4 (B) 0 (C) 2 (D) 1

Q37. JEE Main 2024 (Online) 31st January Morning Shift

Let S be the set of positive integral values of a for which $\frac{ax^2+2(a+1)x+9a+4}{x^2-8x+32} < 0, \forall x \in \mathbb{R}$. Then, the number of elements in S is :

(A) 0 (B) ∞ (C) 3 (D) 1

Q38. JEE Main 2024 (Online) 27th January Evening Shift

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then :

(A) $2S_{12} = S_{11} + S_{10}$ (B) $S_{12} = S_{11} + S_{10}$ (C) $S_{11} = S_{10} + S_{12}$ (D) $2S_{11} = S_{12} + S_{10}$

Q39. JEE Main 2023 (Online) 15th April Morning Shift

The number of real roots of the equation $x|x| - 5|x+2| + 6 = 0$, is :

(A) 4 (B) 3 (C) 5 (D) 6

Q40. JEE Main 2023 (Online) 13th April Evening Shift

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to

(A) -64 (B) $-64\sqrt{2}$ (C) $-128\sqrt{2}$ (D) -128

Q41. JEE Main 2023 (Online) 13th April Morning Shift

The set of all $a \in \mathbb{R}$ for which the equation $x|x-1| + |x+2| + a = 0$ has exactly one real root, is :

(A) $(-\infty, \infty)$ (B) $(-6, \infty)$ (C) $(-\infty, -3)$ (D) $(-6, -3)$

Q42. JEE Main 2023 (Online) 12th April Morning Shift

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ is equal to :

(A) 72 (B) 9 (C) 729 (D) 81

Q43. JEE Main 2023 (Online) 8th April Morning Shift

Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$. If $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to :

(A) 21 (B) 19 (C) $\frac{169}{8}$ (D) $\frac{155}{8}$

Q44. JEE Main 2023 (Online) 6th April Morning Shift

Let $A = \{x \in R : [x + 3] + [x + 4] \leq 3\}$,

$B = \left\{x \in R : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$, where [t] denotes greatest integer function. Then,

(A) $B \subset C, A \neq B$ (B) $A \subset B, A \neq B$ (C) $A = B$ (D) $A \cap B = \phi$

Q45. JEE Main 2023 (Online) 6th April Morning Shift

The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is :

(A) $11 + \sqrt{3}$ (B) $9 + \sqrt{3}$ (C) $9 - \sqrt{3}$ (D) $11 - \sqrt{3}$

Q46. JEE Main 2023 (Online) 1st February Evening Shift

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1, 2) and its other root lies in the interval (2, 3), is :

(A) 2 (B) 0 (C) 1 (D) 3

Q47. JEE Main 2023 (Online) 1st February Morning Shift

Let $S = \left\{x : x \in \mathbb{R} \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10\right\}$. Then $n(S)$ is equal to

(A) 6 (B) 4 (C) 0 (D) 2

Q48. JEE Main 2023 (Online) 31st January Evening Shift

The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R}$ has :

(A) two solutions and both are negative

(B) two solutions and only one of them is negative

(C) four solutions two of which are negative

(D) no solution

Q49. JEE Main 2023 (Online) 31st January Morning Shift

The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is :

(A) 0 (B) 1 (C) 3 (D) 2

Q50. JEE Main 2023 (Online) 29th January Morning Shift

Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

(A) $7x^2 - 245x + 250 = 0$

(B) $49x^2 - 245x + 250 = 0$

(C) $49x^2 + 245x + 250 = 0$

(D) $7x^2 + 245x - 250 = 0$

Q51. JEE Main 2023 (Online) 24th January Evening Shift

The number of real solutions of the equation $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$, is

(A) 3

(B) 4

(C) 0

(D) 2

Q52. JEE Main 2023 (Online) 24th January Morning Shift

The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has :

(A) exactly two solutions in $(-\infty, \infty)$

(B) no solution

(C) a unique solution in $(-\infty, \infty)$

(D) a unique solution in $(-\infty, 1)$

Q53. JEE Main 2022 (Online) 28th July Evening Shift

Let $S = \left\{x \in [-6, 3] - \{-2, 2\} : \frac{|x+3|-1}{|x|-2} \geq 0\right\}$ and $T = \left\{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\right\}$.

Then the number of elements in $S \cap T$ is :

(A) 7

(B) 5

(C) 4

(D) 3

Q54. JEE Main 2022 (Online) 28th July Evening Shift

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are :

(A) non-real complex numbers

(B) real and both negative

(C) real and both positive

(D) real and exactly one of them is positive

Q55. JEE Main 2022 (Online) 27th July Evening Shift

If α, β are the roots of the equation

$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0,$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is :

(A) $3x^2 - 20x - 12 = 0$

(B) $3x^2 - 10x - 4 = 0$

(C) $3x^2 - 10x + 2 = 0$

(D) $3x^2 - 20x + 16 = 0$

Q56. JEE Main 2022 (Online) 26th July Evening Shift

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

(A) 4 (B) 5 (C) 6 (D) 8

Q57. JEE Main 2022 (Online) 25th July Morning Shift

If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to :

(A) -4 (B) -1 (C) 1 (D) 4

Q58. JEE Main 2022 (Online) 30th June Morning Shift

Let $S_1 = \left\{x \in R - \{1, 2\} : \frac{(x+2)(x^2+3x+5)}{-2+3x-x^2} \geq 0\right\}$ and $S_2 = \left\{x \in R : 3^{2x} - 3^{x+1} - 3^{x+2} + 27 \leq 0\right\}$. Then, $S_1 \cup S_2$ is equal to :

(A) $(-\infty, -2] \cup (1, 2)$ (B) $(-\infty, -2] \cup [1, 2]$ (C) $(-2, 1] \cup [2, \infty)$ (D) $(-\infty, 2]$

Q59. JEE Main 2022 (Online) 30th June Morning Shift

Let S be the set of all integral values of α for which the sum of squares of two real roots of the quadratic equation $3x^2 + (\alpha - 6)x + (\alpha + 3) = 0$ is minimum. Then S :

(A) is an empty set (B) is a singleton

(C) contains exactly two elements (D) contains more than two elements

Q60. JEE Main 2022 (Online) 29th June Evening Shift

Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then, the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to :

(A) 1 (B) α (C) $1 + \alpha$ (D) $1 + 2\alpha$

Q61. JEE Main 2022 (Online) 28th June Evening Shift

Let f(x) be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1, then the sum of the roots of $f(x) = 0$ is equal to :

(A) $\frac{11}{3}$ (B) $\frac{7}{3}$ (C) $\frac{13}{3}$ (D) $\frac{14}{3}$

Q62. JEE Main 2022 (Online) 27th June Morning Shift

The number of distinct real roots of $x^4 - 4x + 1 = 0$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Q63. JEE Main 2022 (Online) 25th June Evening Shift

Let $A = \{x \in R : |x + 1| < 2\}$ and $B = \{x \in R : |x - 1| \geq 2\}$. Then which one of the following statements is NOT true?

(A) $A - B = (-1, 1)$

(B) $B - A = R - (-3, 1)$

(C) $A \cap B = (-3, -1]$

(D) $A \cup B = R - [1, 3)$

Q64. JEE Main 2022 (Online) 25th June Evening Shift

Let a, b $\in$ R be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α. If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to :

(A) 37 (B) 58 (C) 68 (D) 92

Q65. JEE Main 2022 (Online) 24th June Evening Shift

The sum of all the real roots of the equation

$(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

(A) $\log_e 3$ (B) $-\log_e 3$ (C) $\log_e 6$ (D) $-\log_e 6$

Q66. JEE Main 2022 (Online) 24th June Evening Shift

The number of distinct real roots of the equation

$x^7 - 7x - 2 = 0$ is

(A) 5 (B) 7 (C) 1 (D) 3

Q67. JEE Main 2022 (Online) 24th June Morning Shift

If the sum of the squares of the reciprocals of the roots α and β of

the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

(A) 18 (B) 24 (C) 36 (D) 96

Q68. JEE Main 2021 (Online) 1st September Evening Shift

The number of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is :

(A) 6 (B) 2 (C) 4 (D) 8

Q69. JEE Main 2021 (Online) 31st August Evening Shift

The sum of the roots of the equation

$x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is :

(A) $\log_2 14$ (B) $\log_2 11$ (C) $\log_2 12$ (D) $\log_2 13$

Q70. JEE Main 2021 (Online) 31st August Morning Shift

cosec18° is a root of the equation :

(A) $x^2 + 2x - 4 = 0$ (B) $4x^2 + 2x - 1 = 0$ (C) $x^2 - 2x + 4 = 0$ (D) $x^2 - 2x - 4 = 0$

Q71. JEE Main 2021 (Online) 27th August Evening Shift

The set of all values of $k > -1$, for which the equation
$(3x^2 + 4x + 3)^2 - (k+1)(3x^2 + 4x + 3)(3x^2 + 4x + 2) + k(3x^2 + 4x + 2)^2 = 0$ has real roots, is :

(A) $\left(1, \frac{5}{2}\right]$ (B) [2, 3)

(C) $\left[-\frac{1}{2}, 1\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right] - \{1\}$

Q72. JEE Main 2021 (Online) 27th July Evening Shift

Let $\alpha = \max_{x \in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$ and $\beta = \min_{x \in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$. If $8x^2 + bx + c = 0$ is a quadratic equation whose roots are $\alpha^{1/5}$ and $\beta^{1/5}$, then the value of $c - b$ is equal to :

(A) 42 (B) 47 (C) 43 (D) 50

Q73. JEE Main 2021 (Online) 27th July Morning Shift

Let α, β be two roots of the

equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to

(A) 10 (B) 100 (C) 50 (D) 160

Q74. JEE Main 2021 (Online) 25th July Evening Shift

If [x] be the greatest integer less than or equal to x,

then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to :

(A) 0 (B) 4 (C) -2 (D) 2

Q75. JEE Main 2021 (Online) 25th July Evening Shift

The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is :

(A) 2 (B) 3 (C) 1 (D) 4

Q76. JEE Main 2021 (Online) 25th July Morning Shift

The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is :

(A) 2 (B) 4 (C) 6 (D) 1

Q77. JEE Main 2021 (Online) 22th July Evening Shift

Let [x] denote the greatest integer less than or equal to x. Then, the values of $x \in R$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval :

(A) $\left[0, \frac{1}{e}\right)$ (B) $[\log_e 2, \log_e 3)$ (C) [1, e) (D) $[0, \log_e 2)$

Q78. JEE Main 2021 (Online) 20th July Morning Shift

If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to :

(A) 56×3^{25} (B) 56×3^{24} (C) 52×3^{24} (D) 28×3^{25}

Q79. JEE Main 2021 (Online) 18th March Morning Shift

The value of $3 + \cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+....\infty}}}}$ is equal to

(A) 1.5 + $\sqrt{3}$ (B) 2 + $\sqrt{3}$ (C) 3 + 2$\sqrt{3}$ (D) 4 + $\sqrt{3}$

Q80. JEE Main 2021 (Online) 17th March Morning Shift

The value of $4 + \cfrac{1}{5+\cfrac{1}{4+\cfrac{1}{5+\cfrac{1}{4+.....\infty}}}}$ is :

(A) 2 + $\frac{2}{5}\sqrt{30}$ (B) 2 + $\frac{4}{\sqrt{5}}\sqrt{30}$

(C) 5 + $\frac{2}{5}\sqrt{30}$ (D) 4 + $\frac{4}{\sqrt{5}}\sqrt{30}$

Q81. JEE Main 2021 (Online) 16th March Morning Shift

The number of elements in the set $\{x \in R : (|x| - 3)|x + 4| = 6\}$ is equal to :

(A) 4 (B) 2 (C) 3 (D) 1

Q82. JEE Main 2021 (Online) 25th February Evening Shift

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10}-2a_8}{3a_9}$ is :

(A) 3 (B) 2 (C) 4 (D) 1

Q83. JEE Main 2021 (Online) 25th February Morning Shift

The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R, is :

(A) 4 (B) 2 (C) 3 (D) 0

Q84. JEE Main 2021 (Online) 24th February Morning Shift

Let p and q be two positive numbers such that p + q = 2 and $p^4+q^4 = 272$. Then p and q are roots of the equation :

(A) $x^2 - 2x + 8 = 0$ (B) $x^2 - 2x + 136 = 0$ (C) $x^2 - 2x + 16 = 0$ (D) $x^2 - 2x + 2 = 0$

Q85. JEE Main 2020 (Online) 6th September Evening Slot

If α and β are the roots of the equation $2x(2x + 1) = 1$, then β is equal to :

(A) $-2\alpha(\alpha + 1)$ (B) $2\alpha(\alpha + 1)$

(C) $2\alpha^2$ (D) $2\alpha(\alpha - 1)$

Q86. JEE Main 2020 (Online) 6th September Morning Slot

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$ is :

(A) 1 (B) 3 (C) 2 (D) 4

Q87. JEE Main 2020 (Online) 5th September Evening Slot

If α and β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to :

(A) $\frac{1}{24}$ (B) $\frac{27}{32}$ (C) $\frac{27}{16}$ (D) $\frac{3}{8}$

Q88. JEE Main 2020 (Online) 5th September Morning Slot

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$ is :

(A) $\frac{5}{9}$ (B) $\frac{5}{27}$ (C) $\frac{25}{81}$ (D) $\frac{25}{9}$

Q89. JEE Main 2020 (Online) 4th September Evening Slot

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

(A) 36 (B) 9 (C) 27 (D) 18

Q90. JEE Main 2020 (Online) 4th September Morning Slot

Let [t] denote the greatest integer $\leq$ t. Then the equation in x, $[x]^2 + 2[x+2] - 7 = 0$ has :

(A) no integral solution.

(B) exactly two solutions.

(C) exactly four integral solutions.

(D) infinitely many solutions.

Q91. JEE Main 2020 (Online) 4th September Morning Slot

Let α and β be the roots of $x^2 - 3x + p=0$ and γ and δ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ form a geometric progression.Then ratio (2q + p) : (2q - p) is:

(A) 9 : 7 (B) 5 : 3 (C) 3 : 1 (D) 33 :31

Q92. JEE Main 2020 (Online) 3rd September Evening Slot

The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval (0, 1) is :

(A) (−3, −1) (B) (2, 4] (C) (0, 2) (D) (1, 3]

Q93. JEE Main 2020 (Online) 3rd September Morning Slot

If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to :

(A) $\frac{9}{4}\left(9 - q^2\right)$ (B) $\frac{9}{4}\left(9 + q^2\right)$

(C) $\frac{9}{4}\left(9 - p^2\right)$ (D) $\frac{9}{4}\left(9 + p^2\right)$

Q94. JEE Main 2020 (Online) 2nd September Evening Slot

Let f(x) be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in :

(A) (−3, −1) (B) (1, 3) (C) (−1, 0) (D) (0, 1)

Q95. JEE Main 2020 (Online) 2nd September Morning Slot

Let α and β be the roots of the equation
$5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, n = 1, 2, 3...., then :

(A) $5S_6 + 6S_5 = 2S_4$

(B) $5S_6 + 6S_5 + 2S_4 = 0$

(C) $6S_6 + 5S_5 + 2S_4 = 0$

(D) $6S_6 + 5S_5 = 2S_4$

Q96. JEE Main 2020 (Online) 9th January Evening Slot

Let a, b $\in$ R, a $\neq$ 0 be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to :

(A) 28 (B) 24 (C) 26 (D) 25

Q97. JEE Main 2020 (Online) 9th January Morning Slot

The number of real roots of the equation,
$e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is :

(A) 1 (B) 2 (C) 3 (D) 4

Q98. JEE Main 2020 (Online) 8th January Evening Slot

Let S be the set of all real roots of the equation,
$3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S :

(A) contains exactly two elements.

(B) is an empty set.

(C) is a singleton.

(D) contains at least four elements.

Q99. JEE Main 2020 (Online) 8th January Evening Slot

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$.
If $a = (1+\alpha)\sum_{k=0}^{100} \alpha^{2k}$ and
$b = \sum_{k=0}^{100} \alpha^{3k}$, then a and b are the roots of the quadratic equation :

(A) $x^2 + 101x + 100 = 0$

(B) $x^2 + 102x + 101 = 0$

(C) $x^2 - 102x + 101 = 0$

(D) $x^2 - 101x + 100 = 0$

Q100. JEE Main 2020 (Online) 7th January Evening Slot

Let α and β be the roots of the equation x^2 - x - 1 = 0.
If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

(A) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(B) $p_5 = 11$

(C) $p_3 = p_5 - p_4$

(D) $p_5 = p_2 \cdot p_3$

Q101. JEE Main 2020 (Online) 7th January Morning Slot

Let α and β be two real roots of the equation
$(k + 1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1 - k)$, where $k(\neq -1)$ and λ are real numbers. if $\tan^2(\alpha + \beta) = 50$, then a value of λ is:

(A) $5\sqrt{2}$ (B) 10 (C) 5 (D) $10\sqrt{2}$

Q102. JEE Main 2019 (Online) 12th April Evening Slot

If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to :

(A) $\alpha\gamma$ (B) 0 (C) $\beta\gamma$ (D) $\alpha\beta$

Q103. JEE Main 2019 (Online) 10th April Evening Slot

The number of real roots of the equation
$5 + |2^x - 1| = 2^x(2^x - 2)$ is

(A) 2 (B) 1 (C) 3 (D) 4

Q104. JEE Main 2019 (Online) 10th April Morning Slot

All the pairs (x, y) that satisfy the inequality
$2^{\sqrt{\sin^2 x - 2\sin x + 5}} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$
also satisfy the equation

(A) sin x = |sin y| (B) sin x = 2sin y
(C) 2 sin x = sin y (D) 2 |sin x | = 3 sin y

Q105. JEE Main 2019 (Online) 10th April Morning Slot

If α and β are the roots of the quadratic equation,
$x^2 + x \sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then
$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}).(\alpha - \beta)^{24}}$ is equal to :

(A) $\frac{2^{12}}{(\sin\theta - 8)^6}$ (B) $\frac{2^6}{(\sin\theta + 4)^{12}}$
(C) $\frac{2^{12}}{(\sin\theta + 8)^{12}}$ (D) $\frac{2^{12}}{(\sin\theta - 4)^{12}}$

Q106. JEE Main 2019 (Online) 9th April Evening Slot

If m is chosen in the quadratic equation

$(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$

such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is :-

(A) $4\sqrt{3}$ (B) $8\sqrt{3}$ (C) $8\sqrt{5}$ (D) $10\sqrt{5}$

Q107. JEE Main 2019 (Online) 9th April Morning Slot

Let p, q $\in$ R. If 2 - $\sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then :

(A) $p^2 - 4q - 12 = 0$ (B) $q^2 - 4p - 16 = 0$ (C) $q^2 + 4p + 14 = 0$ (D) $p^2 - 4q + 12 = 0$

Q108. JEE Main 2019 (Online) 8th April Evening Slot

The number of integral values of m for which the equation

$(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :

(A) 2 (B) infinitely many (C) 1 (D) 3

Q109. JEE Main 2019 (Online) 8th April Morning Slot

The sum of the solutions of the equation
$\left|\sqrt{x}-2\right|+\sqrt{x}\left(\sqrt{x}-4\right)+2=0$
(x > 0) is equal to:

(A) 9 (B) 12 (C) 4 (D) 10

Q110. JEE Main 2019 (Online) 12th January Evening Slot

The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, x $\in$ R, is always positive, is :

(A) 7 (B) 8 (C) 3 (D) 6

Q111. JEE Main 2019 (Online) 12th January Morning Slot

If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda+\frac{1}{\lambda}=1$, is

(A) $-2+\sqrt{2}$ (B) $4-3\sqrt{2}$ (C) $2-\sqrt{3}$ (D) $4-2\sqrt{3}$

Q112. JEE Main 2019 (Online) 11th January Evening Slot

Let α and β be the roots of the quadratic equation $x^2 \sin\theta - x(\sin\theta\cos\theta + 1) + \cos\theta = 0$ ($0 < \theta < 45^o$), and $\alpha < \beta$. Then $\sum_{n=0}^{\infty}\left(\alpha^n+\frac{(-1)^n}{\beta^n}\right)$ is equal to :

(A) $\frac{1}{1+\cos\theta}+\frac{1}{1-\sin\theta}$ (B) $\frac{1}{1-\cos\theta}+\frac{1}{1+\sin\theta}$

(C) $\frac{1}{1-\cos\theta}-\frac{1}{1+\sin\theta}$ (D) $\frac{1}{1+\cos\theta}-\frac{1}{1-\sin\theta}$

Q113. JEE Main 2019 (Online) 11th January Morning Slot

If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube of the other root, then a value of k is

(A) -81 (B) -300 (C) 100 (D) 144

Q114. JEE Main 2019 (Online) 10th January Evening Slot

The value of λ such that sum of the squares of the roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is -

(A) 1 (B) 2 (C) $\frac{15}{8}$ (D) $\frac{4}{9}$

Q115. JEE Main 2019 (Online) 10th January Morning Slot

If 5, 5r, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to :

(A) $\frac{7}{4}$ (B) $\frac{5}{4}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Q116. JEE Main 2019 (Online) 10th January Morning Slot

Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0, 2) and its other root lies in the interval (2, 3). Then the number of elements in S is -

(A) 12 (B) 18 (C) 10 (D) 11

Q117. JEE Main 2019 (Online) 9th January Evening Slot

If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval [1, 5], then m lies in the interval :

(A) $(-5, -4)$ (B) (4, 5) (C) (5, 6) (D) (3, 4)

Q118. JEE Main 2019 (Online) 9th January Evening Slot

The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is :

(A) 3 (B) 2 (C) 4 (D) 5

Q119. JEE Main 2018 (Online) 16th April Morning Slot

Let p, q and r be real numbers ($p \neq q$, $r \neq 0$), such that the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the sum of squares of these roots is equal to :

(A) $\frac{p^2+q^2}{2}$ (B) $p^2 + q^2$ (C) $2(p^2 + q^2)$ (D) $p^2 + q^2 + r^2$

Q120. JEE Main 2018 (Online) 16th April Morning Slot

If an angle A of a ΔABC satiesfies $5\cos A + 3 = 0$, then the roots of the quadratic equation, $9x^2 + 27x + 20 = 0$ are :

(A) secA, cotA (B) sinA, secA (C) secA, tanA (D) tanA, cosA

Q121. JEE Main 2018 (Offline)

Let S = { $x \in$ R : $x \geq 0$ and

$2\left|\sqrt{x}-3\right|+\sqrt{x}\left(\sqrt{x}-6\right)+6=0$}. Then S

(A) contains exactly four elements

(B) is an empty set

(C) contains exactly one element

(D) contains exactly two elements

Q122. JEE Main 2018 (Online) 15th April Evening Slot

If f(x) is a quadratic expression such that f (1) + f (2) = 0, and -1 is a root of f (x) = 0, then the other root of f(x) = 0 is :

(A) $-\frac{5}{8}$ (B) $-\frac{8}{5}$ (C) $\frac{5}{8}$ (D) $\frac{8}{5}$

Q123. JEE Main 2018 (Online) 15th April Morning Slot

If $\lambda \in$ **R** is such that the sum of the cubes of the roots of the equation,
$x^2 + (2-\lambda)x + (10-\lambda) = 0$ is minimum, then the magnitude of the difference of the roots of this equation is :

(A) $4\sqrt{2}$ (B) $2\sqrt{5}$ (C) $2\sqrt{7}$ (D) 20

Q124. JEE Main 2018 (Online) 15th April Morning Slot

If tanA and tanB are the roots of the quadratic equation, $3x^2 - 10x - 25 = 0$, then the value of $3\sin^2(A+B) - 10\sin(A+B).\cos(A+B) - 25\cos^2(A+B)$ is :

(A) -10 (B) 10 (C) -25 (D) 25

Q125. JEE Main 2017 (Online) 9th April Morning Slot

The sum of all the real values of x satisfying the equation
$2^{(x-1)(x^2+5x-50)} = 1$ is :

(A) 16 (B) 14 (C) -4 (D) -5

Q126. JEE Main 2017 (Online) 8th April Morning Slot

Let p(x) be a quadratic polynomial such that p(0)=1. If p(x) leaves remainder 4 when divided by $x-1$ and it leaves remainder 6 when divided by x + 1; then :

(A) p(2) = 11 (B) p(2) = 19 (C) p(-2) = 19 (D) p(-2) = 11

Q127. JEE Main 2017 (Offline)

If for a positive integer n, the quadratic equation

$$x(x+1)+(x+1)(x+2)+\ldots+\left(x+\overline{n-1}\right)(x+n)=10n$$

has two consecutive integral solutions, then n is equal to :

(A) 9 (B) 10 (C) 11 (D) 12

Q128. JEE Main 2016 (Online) 10th April Morning Slot

If x is a solution of the equation, $\sqrt{2x+1}-\sqrt{2x-1}=1,\ \left(x\geq\frac{1}{2}\right)$, then $\sqrt{4x^2-1}$ is equal to :

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) $2\sqrt{2}$

Q129. JEE Main 2016 (Online) 9th April Morning Slot

If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from −1, then $|b|$ is equal to :

(A) $\sqrt{2}$ (B) 2 (C) 3 (D) $\sqrt{3}$

Q130. JEE Main 2016 (Offline)

The sum of all real values of x satisfying the equation $\left(x^2-5x+5\right)^{x^2+4x-60}=1$ is :

(A) 6 (B) 5 (C) 3 (D) -4

Q131. JEE Main 2015 (Offline)

Let α and β be the roots of equation $x^2-6x-2=0$. If $a_n=\alpha^n-\beta^n$, for $n\geq 1$, then the value of $\frac{a_{10}-2a_8}{2a_9}$ is equal to :

(A) 3 (B) -3 (C) 6 (D) -6

Q132. JEE Main 2014 (Offline)

If $a\in R$ and the equation $-3(x-[x])^2+2(x-[x])+a^2=0$ (where $[x]$ denotes the greater integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

(A) $(-2,-1)$ (B) $(-\infty,-2)\cup(2,\infty)$

(C) $(-1,0)\cup(0,1)$ (D) $(1,2)$

Q133. JEE Main 2014 (Offline)

Let α and β be the roots of equation $px^2+qx+r=0,\ p\neq 0$. If $p,\ q,\ r$ in A.P. and $\frac{1}{\alpha}+\frac{1}{\beta}=4$, then the value of $|\alpha-\beta|$ is :

(A) $\frac{\sqrt{34}}{9}$ (B) $\frac{2\sqrt{13}}{9}$ (C) $\frac{\sqrt{61}}{9}$ (D) $\frac{2\sqrt{17}}{9}$

Q134. JEE Main 2013 (Offline)

If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a,\ b,\ c \in R$, have a common root, then $a : b : c$ is

(A) $1 : 2 : 3$ (B) $3 : 2 : 1$ (C) $1 : 3 : 2$ (D) $3 : 1 : 2$

Q135. AIEEE 2012

The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has:

(A) infinite number of real roots (B) no real roots

(C) exactly one real root (D) exactly four real roots

Q136. AIEEE 2010

If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$

(A) -1 (B) 1 (C) 2 (D) -2

Q137. AIEEE 2009

If the roots of the equation $bx^2 + cx + a = 0$ are imaginary, then for all real values of x, the expression $3b^2x^2 + 6bcx + 2c^2$ is :

(A) less than $4ab$ (B) greater than $-4ab$ (C) less than $-4ab$ (D) greater than $4ab$

Q138. AIEEE 2008

STATEMENT - 1 : For every natural number $n \geq 2$,

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \ldots\ldots\ldots + \frac{1}{\sqrt{n}} > \sqrt{n}.$$

STATEMENT - 2 : For every natural number $n \geq 2$,,

$$\sqrt{n\,(n+1)} < n + 1.$$

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Q139. AIEEE 2008

The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then the common root is

(A) 1 (B) 4 (C) 3 (D) 2

Q140. AIEEE 2007

If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is

(A) $(3, \infty)$ (B) $(-\infty, -3)$ (C) $(-3, 3)$ (D) $(-3, \infty)$

Q141. AIEEE 2006

If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then the value of $2 + q - p$ is

(A) 2 (B) 3 (C) 0 (D) 1

Q142. AIEEE 2006

All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4, lie in the interval

(A) $-2 < m < 0$ (B) $m > 3$ (C) $-1 < m < 3$ (D) $1 < m < 4$

Q143. AIEEE 2006

If x is real, the maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$ is

(A) $\frac{1}{4}$ (B) 41 (C) 1 (D) $\frac{17}{7}$

Q144. AIEEE 2005

In a triangle PQR, $\angle R = \frac{\pi}{2}$. $If\ \tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0,\ a \neq 0$ then

(A) $a = b + c$ (B) $c = a + b$ (C) $b = c$ (D) $b = a + c$

Q145. AIEEE 2005

If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval

(A) $(5, 6]$ (B) $(6, \infty)$ (C) $(-\infty, 4)$ (D) $[4, 5]$

Q146. AIEEE 2005

The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assumes the least value is

(A) 1 (B) 0 (C) 3 (D) 2

Q147. AIEEE 2005

If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

(A) -2 (B) 3 (C) 2 (D) 1

Q148. AIEEE 2005

The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assumes the least value is :

(A) 1 (B) 0 (C) 3 (D) 2

Q149. AIEEE 2005

If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

(A) -2 (B) 3 (C) 2 (D) 1

Q150. AIEEE 2004

Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

(A) $x^2 - 18x - 16 = 0$ (B) $x^2 - 18x + 16 = 0$ (C) $x^2 + 18x - 16 = 0$ (D) $x^2 + 18x + 16 = 0$

Q151. AIEEE 2004

If $(1-p)$ is a root of quadratic equation $x^2 + px + (1-p) = 0$ then its roots are

(A) $-1, 2$ (B) $-1, 1$ (C) $0, -1$ (D) $0, 1$

Q152. AIEEE 2004

If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of $'q'$ is

(A) 4 (B) 12 (C) 3 (D) $\frac{49}{4}$

Q153. AIEEE 2003

If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}$, $\frac{b}{a}$ and $\frac{c}{b}$ are in

(A) Arithmetic - Geometric Progression (B) Arithmetic Progression

(C) Geometric Progression (D) Harmonic Progression

Q154. AIEEE 2003

The value of 'a' for which one root of the quadratic equation $(a^2-5a+3)x^2+(3a-1)x+2=0$ is twice as large as the other is

(A) $-\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{1}{3}$

Q155. AIEEE 2003

The number of real solutions of the equation $x^2-3|x|+2=0$ is

(A) 3 (B) 2 (C) 4 (D) 1

Q156. AIEEE 2002

If $\alpha \neq \beta$ but $\alpha^2=5\alpha-3$ and $\beta^2=5\beta-3$ then the equation having α/β and β/α as its roots is

(A) $3x^2-19x+3=0$ (B) $3x^2+19x-3=0$ (C) $3x^2-19x-3=0$ (D) $x^2-5x+3=0$

Q157. AIEEE 2002

Difference between the corresponding roots of $x^2+ax+b=0$ and $x^2+bx+a=0$ is same and $a \neq b$, then

(A) $a+b+4=0$ (B) $a+b-4=0$ (C) $a-b-4=0$ (D) $a-b+4=0$

Q158. AIEEE 2002

Product of real roots of equation $t^2x^2+|x|+9=0$

(A) is always positive (B) is always negative
(C) does not exist (D) none of these

Q159. AIEEE 2002

If p and q are the roots of the equation $x^2+px+q=0$, then

(A) $p=1,\ q=-2$ (B) $p=0,\ q=1$ (C) $p=-2,\ q=0$ (D) $p=-2,\ q=1$

Q160. AIEEE 2002

If $a,\ b,\ c$ are distinct $+ve$ real numbers and $a^2+b^2+c^2=1$ then $ab+bc+ca$ is

(A) less than 1 (B) equal to 1 (C) greater than 1 (D) any real no.

Numerical 38 questions

Q161. JEE Main 2025 (Online) 2nd April Evening Shift

If the set of all $\mathrm{a} \in \mathbf{R}-\{1\}$, for which the roots of the equation $(1-\mathrm{a})x^2+2(\mathrm{a}-3)x+9=0$ are positive is $(-\infty,-\alpha] \cup [\beta,\gamma)$, then $2\alpha+\beta+\gamma$ is equal to .

Q162. JEE Main 2025 (Online) 23rd January Morning Shift

If the equation $\mathrm{a}(\mathrm{b}-\mathrm{c})\mathrm{x}^2+\mathrm{b}(\mathrm{c}-\mathrm{a})\mathrm{x}+\mathrm{c}(\mathrm{a}-\mathrm{b})=0$ has equal roots, where $\mathrm{a}+\mathrm{c}=15$ and $\mathrm{b}=\frac{36}{5}$, then a^2+c^2 is equal to _____

Q163. JEE Main 2024 (Online) 8th April Evening Shift

The number of distinct real roots of the equation $|x+1||x+3|-4|x+2|+5=0$, is _____

Q164. JEE Main 2024 (Online) 6th April Evening Shift

Let α, β be roots of $x^2+\sqrt{2}x-8=0$. If $\mathrm{U_n}=\alpha^{\mathrm{n}}+\beta^{\mathrm{n}}$, then $\frac{\mathrm{U_{10}}+\sqrt{2}\mathrm{U_9}}{2\mathrm{U_8}}$ is equal to _____.

Q165. JEE Main 2024 (Online) 6th April Morning Shift

Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4+8x^3-17x^2-12x+9=0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right)=\frac{125}{16}m$. Then the value of m is _____.

Q166. JEE Main 2024 (Online) 5th April Evening Shift

The number of real solutions of the equation $x|x+5|+2|x+7|-2=0$ is _____.

Q167. JEE Main 2024 (Online) 5th April Morning Shift

The number of distinct real roots of the equation $|x||x+2|-5|x+1|-1=0$ is _____.

Q168. JEE Main 2024 (Online) 31st January Evening Shift

Let a, b, c be the lengths of three sides of a triangle satisfying the condition $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$. If the set of all possible values of x is the interval (α, β), then $12\left(\alpha^2+\beta^2\right)$ is equal to _____.

Q169. JEE Main 2024 (Online) 30th January Evening Shift

The number of real solutions of the equation $x\left(x^2+3|x|+5|x-1|+6|x-2|\right)=0$ is _____.

Q170. JEE Main 2024 (Online) 30th January Morning Shift

Let $\alpha, \beta \in \mathbf{N}$ be roots of the equation $x^2-70x+\lambda=0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin \mathbf{N}$. If λ assumes the minimum possible value, then $\frac{(\sqrt{\alpha-1}+\sqrt{\beta-1})(\lambda+35)}{|\alpha-\beta|}$ is equal to :

Q171. JEE Main 2024 (Online) 29th January Evening Shift

Let the set $C=\left\{(x, y) \mid x^2-2^y=2023, x, y \in \mathbb{N}\right\}$. Then $\sum\limits_{(x, y) \in C}(x+y)$ is equal to _____.

Q172. JEE Main 2023 (Online) 13th April Evening Shift

Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $[\sqrt{1}]+[\sqrt{2}]+[\sqrt{3}]+\ldots+[\sqrt{120}]$ is equal to _____

Q173. JEE Main 2023 (Online) 11th April Evening Shift

The number of points, where the curve $f(x)=\mathrm{e}^{8x}-\mathrm{e}^{6x}-3\mathrm{e}^{4x}-\mathrm{e}^{2x}+1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

Q174. JEE Main 2023 (Online) 11th April Morning Shift

If a and b are the roots of the equation $x^2-7x-1=0$, then the value of $\frac{a^{21}+b^{21}+a^{17}+b^{17}}{a^{19}+b^{19}}$ is equal to _____.

Q175. JEE Main 2023 (Online) 8th April Evening Shift

Let m and n be the numbers of real roots of the quadratic equations $x^2-12x+[x]+31=0$ and $x^2-5|x+2|-4=0$ respectively, where $[x]$ denotes the greatest integer $\leq x$. Then $\mathrm{m}^2+\mathrm{mn}+\mathrm{n}^2$ is equal to _____.

Q176. JEE Main 2023 (Online) 30th January Evening Shift

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$

have a common real root is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____.

Q177. JEE Main 2023 (Online) 29th January Evening Shift

Let $\alpha_1, \alpha_2, \ldots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq \ldots \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Q178. JEE Main 2023 (Online) 25th January Evening Shift

Let $a \in \mathbb{R}$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is _____.

Q179. JEE Main 2023 (Online) 24th January Morning Shift

Let $\lambda \in \mathbb{R}$ and let the equation E be $|x|^2 - 2|x| + |\lambda - 3| = 0$. Then the largest element in the set S = $\{x + \lambda : x$ is an integer solution of E$\}$ is _____

Q180. JEE Main 2022 (Online) 29th July Evening Shift

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $n \in \mathrm{N}$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____.

Q181. JEE Main 2022 (Online) 28th July Morning Shift

The sum of all real values of x for which $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ is equal to _____.

Q182. JEE Main 2022 (Online) 26th July Morning Shift

If for some $\mathrm{p}, \mathrm{q}, \mathrm{r} \in \mathbf{R}$, not all have same sign, one of the roots of the equation $\left(\mathrm{p}^2 + \mathrm{q}^2\right)x^2 - 2\mathrm{q}(\mathrm{p} + \mathrm{r})x + \mathrm{q}^2 + \mathrm{r}^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{\mathrm{q}^2 + \mathrm{r}^2}{\mathrm{p}^2}$ is equal to _____,

Q183. JEE Main 2022 (Online) 26th July Morning Shift

The number of distinct real roots of the equation $x^5\left(x^3 - x^2 - x + 1\right) + x\left(3x^3 - 4x^2 - 2x + 4\right) - 1 = 0$ is _____.

Q184. JEE Main 2022 (Online) 28th June Morning Shift

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is _____.

Q185. JEE Main 2022 (Online) 27th June Evening Shift

Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - \left(3\sqrt{2} + 2\sqrt{3}\right)x + 7 + 3\lambda\sqrt{3} = 0$, $\lambda > 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

Q186. JEE Main 2022 (Online) 27th June Morning Shift

If the sum of all the roots of the equation

$e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.

Q187. JEE Main 2022 (Online) 26th June Evening Shift

Let p and q be two real numbers such that p + q = 3 and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

Q188. JEE Main 2022 (Online) 26th June Morning Shift

The sum of the cubes of all the roots of the equation

$x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is _____.

Q189. JEE Main 2021 (Online) 1st September Evening Shift

Let f(x) be a polynomial of degree 3 such that
$f(k) = -\frac{2}{k}$ for k = 2, 3, 4, 5. Then the value of $52 - 10f(10)$ is equal to :

Q190. JEE Main 2021 (Online) 26th August Evening Shift

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$, and α and γ are the roots of equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to _____.

Q191. JEE Main 2021 (Online) 26th August Morning Shift

The sum of all integral values of k ($k \neq 0$) for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is _____.

Q192. JEE Main 2021 (Online) 27th July Evening Shift

The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^{x} + 1 = 0$ is equal to _____.

Q193. JEE Main 2021 (Online) 25th July Evening Shift

If a + b + c = 1, ab + bc + ca = 2 and abc = 3, then the value of $a^4 + b^4 + c^4$ is equal to _____.

Q194. JEE Main 2021 (Online) 25th July Morning Shift

If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to _____.

Q195. JEE Main 2021 (Online) 26th February Evening Shift

Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

Q196. JEE Main 2021 (Online) 26th February Morning Shift

The sum of 162th power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is _____.

Q197. JEE Main 2021 (Online) 24th February Evening Shift

The number of the real roots of the equation $(x+1)^2 + |x-5| = \frac{27}{4}$ is _____.

Q198. JEE Main 2020 (Online) 8th January Morning Slot

The least positive value of 'a' for which the equation

$2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **B**	4. **D**	5. **C**
6. **B**	7. **D**	8. **A**	9. **A**	10. **B**
11. **A**	12. **A**	13. **D**	14. **B**	15. **C**
16. **C**	17. **D**	18. **A**	19. **A**	20. **A**
21. **C**	22. **A**	23. **A**	24. **D**	25. **C**
26. **A**	27. **A**	28. **D**	29. **C**	30. **A**
31. **B**	32. **A**	33. **B**	34. **A**	35. **D**
36. **C**	37. **A**	38. **B**	39. **B**	40. **D**
41. **A**	42. **D**	43. **B**	44. **C**	45. **B**
46. **C**	47. **B**	48. **A**	49. **B**	50. **B**
51. **C**	52. **C**	53. **D**	54. **B**	55. **B**
56. **C**	57. **B**	58. **B**	59. **A**	60. **A**
61. **A**	62. **B**	63. **B**	64. **B**	65. **B**
66. **D**	67. **B**	68. **A**	69. **B**	70. **D**
71. **A**	72. **A**	73. **C**	74. **B**	75. **A**
76. **A**	77. **D**	78. **C**	79. **A**	80. **A**
81. **B**	82. **B**	83. **C**	84. **C**	85. **A**
86. **C**	87. **C**	88. **C**	89. **D**	90. **D**
91. **A**	92. **D**	93. **C**	94. **C**	95. **A**
96. **D**	97. **A**	98. **C**	99. **C**	100. **D**
101. **B**	102. **C**	103. **B**	104. **A**	105. **C**
106. **C**	107. **A**	108. **B**	109. **D**	110. **A**
111. **B**	112. **B**	113. **B**	114. **B**	115. **A**
116. **D**	117. **B**	118. **A**	119. **B**	120. **C**
121. **D**	122. **D**	123. **B**	124. **C**	125. **C**
126. **C**	127. **C**	128. **A**	129. **D**	130. **C**
131. **A**	132. **C**	133. **B**	134. **A**	135. **B**
136. **B**	137. **B**	138. **B**	139. **D**	140. **C**
141. **B**	142. **C**	143. **B**	144. **B**	145. **C**
146. **A**	147. **D**	148. **A**	149. **D**	150. **B**

151. **C**	152. **D**	153. **D**	154. **B**	155. **C**
156. **A**	157. **A**	158. **A**	159. **A**	160. **A**

Numerical

161. **7**	162. **117**	163. **2**	164. **4**	165. **221**
166. **3**	167. **3**	168. **36**	169. **1**	170. **60**
171. **46**	172. **825**	173. **2**	174. **51**	175. **9**
176. **13**	177. **9**	178. **45**	179. **5**	180. **16**
181. **6**	182. **272**	183. **3**	184. **2**	185. **98**
186. **45**	187. **4**	188. **36**	189. **26**	190. **18**
191. **66**	192. **2**	193. **13**	194. **1**	195. **324**
196. **3**	197. **2**	198. **8**		